

EVIDENCE · CANVAS · PROSE

math

Boardroom-quality math layouts for mathematicians, quants, ML researchers, physicists, statisticians, and economists. KaTeX-rendered equations with persona-appropriate surround.

The closed-form estimator.

LATTICE • MATH

$$\hat{\beta} = (X^{\top} X)^{-1} X^{\top} y$$

WHERE

- $\hat{\beta}$ — OLS coefficient vector
- X — design matrix, $n \times p$
- y — response vector, length n
- $X^{\top} X$ — Gram matrix, $p \times p$, must be invertible

The log-likelihood we maximise.

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$$\ell(\beta) = \sum_{i=1}^n \left[y_i \log \sigma(x_i^\top \beta) + (1 - y_i) \log(1 - \sigma(x_i^\top \beta)) \right]$$

WHERE

- ℓ — log-likelihood, concave in β
- σ — logistic link, $\sigma(z) = 1/(1 + e^{-z})$
- y_i — observed label, $\in \{0, 1\}$
- x_i — feature vector for observation i

Derivative of f from first principles.

$$f(x + h) = f(x) + f'(x) h + O(h^2)$$

Taylor expansion, $n = 2$

$$f(x + h) - f(x) = f'(x) h + O(h^2)$$

subtract $f(x)$

$$\frac{f(x + h) - f(x)}{h} = f'(x) + O(h)$$

divide by $h \neq 0$

$$\lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = f'(x)$$

take the limit

Intermediate Value Theorem.

DEFINITION. A function $f : [a, b] \rightarrow \mathbb{R}$ is *continuous* on $[a, b]$ if $\lim_{x \rightarrow c} f(x) = f(c)$ for every $c \in [a, b]$.

THEOREM. Let f be continuous on $[a, b]$ and let y lie strictly between $f(a)$ and $f(b)$. Then there exists $c \in (a, b)$ with $f(c) = y$.

PROOF. Set $S = \{x \in [a, b] : f(x) < y\}$. S is non-empty and bounded; let $c = \sup S$. Continuity at c forces $f(c) = y$. $\square$

Frequentist vs Bayesian point estimate.

FREQUENTIST

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(y \mid \theta)$$

Maximises the likelihood — no prior. Uncertainty quantified by the sampling distribution of $\hat{\theta}$ across hypothetical repeats.

BAYESIAN

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid y)$$

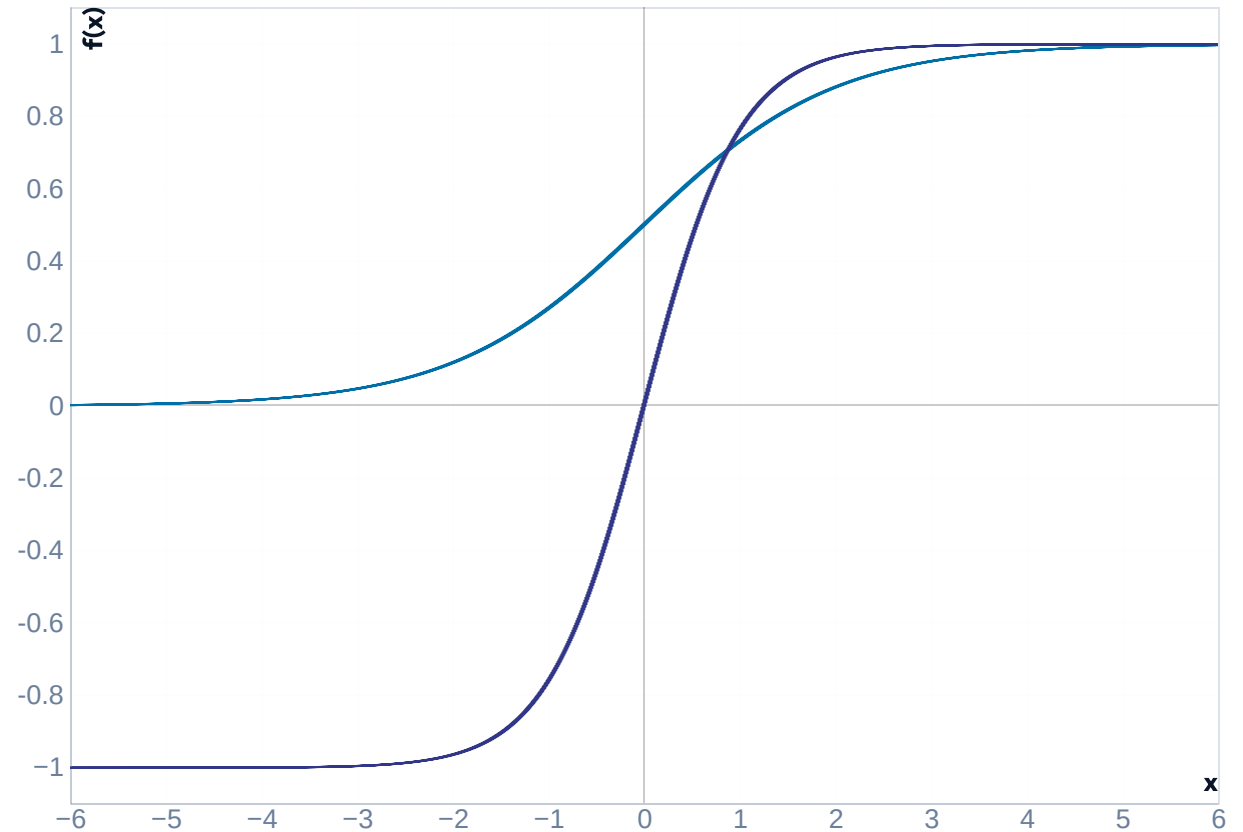
Maximises the posterior — conditions on the prior $p(\theta)$. Uncertainty is the posterior itself, no repeated sampling required.

The sigmoid.

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$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Maps $\mathbb{R} \rightarrow (0, 1)$. *S*-shaped, $\sigma(0) = 0.5$,
steepest slope at the origin.



The design matrix X .

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$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix}$$

- `SHAPE` — $n \times (p + 1)$
- `ROWS` — observations
- `COLS` — intercept + p features
- `RANK` — full-rank for OLS to have a unique solution
- `COLUMN 0` — all-ones, absorbs the intercept

Effect of the treatment.

$$\hat{\beta} = 0.42 \pm 0.03$$

$$\begin{array}{l} 95\% \text{ CI: } [0.36, 0.48] \\ p < 0.001 \quad \cdot \quad n = 1,204 \end{array}$$

*For every additional unit of exposure, the outcome rises by 0.42 SD — roughly an **8%** shift on the baseline. Effect size is the headline; the p -value just rules out chance.*

LU decomposition.

$$\begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

- A — the original matrix being factorised
- L — lower-triangular, unit diagonal
 - U — upper-triangular
- `USE` — solve $Ax = b$ by forward then back substitution

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When NOT to reach for math.

Two display equations in the base layout. The bare math layout is built around one hero equation. For side-by-side display, use

`math`
`compare`

. For a derivation chain, use

`math`
`derivation`

Stacking two

`$$`

blocks in the base layout breaks the visual contract.

Symbols without a legend. An equation with three undefined symbols is a puzzle, not a claim. Either every non-trivial symbol gets a legend entry, or the equation is simple enough that the audience knows it cold.

ASCII math instead of KaTeX.

Writing `beta_hat = (X'X)^-1 X'y` as plain text bypasses the renderer. Always wrap math in

`$$...` (display)
`$$` or

`$...`
`$`

(inline) — KaTeX is the entire reason this layout exists.

RELATED COMPONENTS

See also.

- `code` — the implementation, not the equation, is the argument
- `diagram` — the structure of the model, not its closed form
- `stats` — a row of statistical results without a single equation focus
- `content` — one inline equation inside a paragraph of prose